Chapter 7 — Reaction Kinetics

Board-style MCQs · PECTAA Chemistry 11 · Answer key at end

Practice MCQs

1. Rate of reaction is change in:

(a) Temperature only **(b)** Volume per gas **(c)** Mass per mole **(d)** Concentration per time

2. Rate constant k is independent of:

(a) Temperature **(b)** Nature of reaction **(c)** Catalyst **(d)** Concentration

3. Effective collisions need:

(a) Any collision **(b)** No orientation **(c)** Enough energy and orientation **(d)** Low energy

4. Activation energy is:

(a) Bond length **(b)** Entropy **(c)** Heat of reaction **(d)** Minimum energy for reaction

5. Increasing temperature increases:

(a) Only collision size **(b)** ΔH **(c)** Fraction above Ea **(d)** Activation energy

6. Catalyst increases rate by:

(a) Changing ΔH **(b)** Lowering Ea **(c)** Consuming reactants **(d)** Raising Ea

7. Powdered solid reacts faster due to:

(a) Lower temperature **(b)** No collisions **(c)** More surface area **(d)** Less mass

8. Order of reaction is found by:

(a) Colour **(b)** Experiment **(c)** Balanced equation only **(d)** Molar mass

9. Molecularity applies to:

(a) Catalyst only **(b)** Overall only **(c)** Elementary step **(d)** Product only

10. Zero order means rate is:

(a) Independent of concentration **(b)** Proportional to square **(c)** Zero always **(d)** Proportional to concentration

11. First order plot ln[A] vs t is:

(a) Random **(b)** Circular **(c)** Linear **(d)** Parabolic

12. Rate-determining step is:

(a) Last step always **(b)** Fastest step **(c)** First step always **(d)** Slowest step

13. Increasing pressure increases gas rate by:

(a) Lowering temperature **(b)** Removing catalyst **(c)** Increasing volume **(d)** Increasing concentration

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14. Boltzmann curve shifts at higher T to:

(a) No change **(b)** Lower energy **(c)** Zero energy **(d)** Higher energy

15. A catalyst is:

(a) A product **(b)** Chemically unchanged overall **(c)** A reactant permanently **(d)** Always consumed

16. Rate law for A -> products:

(a) k[A]^n **(b)** k/[A] **(c)** k + [A] **(d)** [A]/k

17. Molecularity cannot be:

(a) Three **(b)** One **(c)** Zero or fraction **(d)** Two

18. Order can be:

(a) Only positive integer **(b)** Zero, integer or fraction **(c)** Only two **(d)** Only one

19. Fast reactions example:

(a) Weathering **(b)** Rusting **(c)** $AgNO_3$ + NaCl **(d)** Fermentation

20. Slow reaction example:

(a) Neutralization **(b)** Explosion **(c)** Precipitation **(d)** Rusting

21. Collision frequency increases with:

(a) Larger container **(b)** Concentration **(c)** Lower temperature **(d)** Less pressure

22. Orientation factor matters for:

(a) All collisions **(b)** Only gases **(c)** No reactions **(d)** Effective collisions

23. Half-life of first order is:

(a) Zero **(b)** Increases with time **(c)** Constant **(d)** Depends on initial mass only

24. Rate unit usually:

(a) mol only **(b)** mol dm^{-3} s^{-1} **(c)** g only **(d)** s only

25. Activated complex is:

(a) Final product **(b)** Solvent **(c)** Catalyst **(d)** High-energy transition state

26. Increasing surface area affects:

(a) Only solutions **(b)** No reactions **(c)** Only gases **(d)** Heterogeneous reactions

27. Rate constant units depend on:

(a) Colour **(b)** Volume only **(c)** Mass **(d)** Order

28. Temperature rise usually rate:

(a) Decreases **(b)** Stops **(c)** Increases greatly **(d)** No change

29. Mechanism must agree with:

(a) Density **(b)** Experimental rate law **(c)** Colour only **(d)** Molar mass

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30. Intermediates are:

(a) Catalysts only **(b)** Final products **(c)** Never formed **(d)** Formed then consumed

31. Collision theory requires:

(a) All collisions **(b)** Effective collisions **(c)** Only slow collisions **(d)** No collisions

32. Effective collision needs:

(a) Only high speed **(b)** E >= Ea and proper orientation **(c)** Only large size **(d)** Low temperature

33. Activation energy is:

(a) Minimum energy for reaction **(b)** Bond length **(c)** Molar mass **(d)** Heat released

34. Rate unit is usually:

(a) $dm^3\ s^{-1}$ **(b)** $mol\ dm^{-3}\ s^{-1}$ **(c)** $mol\ s^{-1}$ **(d)** $g\ mol^{-1}$

35. Average rate is over:

(a) Zero time **(b)** Time interval **(c)** One instant only **(d)** No time

36. Instantaneous rate is from:

(a) Total mass **(b)** Tangent slope **(c)** Colour only **(d)** Initial mass only

37. Higher concentration increases:

(a) Collision frequency **(b)** Bond length **(c)** Activation energy **(d)** Molar mass

38. Higher temperature increases:

(a) Bond length **(b)** Activation energy **(c)** Molar mass **(d)** Fraction above Ea

39. Powdered solid reacts faster due to:

(a) Lower mass **(b)** Surface area **(c)** Lower temperature **(d)** Higher density

40. Catalyst provides:

(a) More heat **(b)** More reactant **(c)** Higher Ea path **(d)** Alternative lower Ea path

41. Boltzmann distribution at higher T:

(a) Disappears **(b)** Becomes vertical **(c)** Narrows left **(d)** Flattens and shifts right

42. Rate law is determined by:

(a) Balancing only **(b)** Colour **(c)** Molar mass **(d)** Experiment

43. Rate constant k changes with:

(a) Pressure only **(b)** Concentration **(c)** Volume only **(d)** Temperature

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44. Overall order is:

(a) Molecularity always **(b)** Sum of powers **(c)** Product of powers **(d)** Number of reactants

45. Zero order means rate is:

(a) Independent of concentration **(b)** Zero always **(c)** Proportional to square **(d)** Proportional to concentration

46. First order doubling concentration:

(a) No change **(b)** Quadruples rate **(c)** Halves rate **(d)** Doubles rate

47. Second order doubling concentration:

(a) No change **(b)** Quadruples rate **(c)** Halves rate **(d)** Doubles rate

48. Molecularity is:

(a) Always fractional **(b)** Experimental order **(c)** Particles in elementary step **(d)** Always zero

49. Rate-determining step is:

(a) First step always **(b)** Fastest step **(c)** Last step always **(d)** Slowest step

50. First-order half-life is:

(a) Decreases with concentration **(b)** Constant **(c)** Increases with concentration **(d)** Zero

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Answer Key with Brief Reasons

1. Answer: (d) Concentration per time — Kinetics definition.

2. Answer: (d) Concentration — k changes with T/catalyst.

3. Answer: (c) Enough energy and orientation — Collision theory.

4. Answer: (d) Minimum energy for reaction — Energy barrier.

5. Answer: (c) Fraction above Ea — Boltzmann distribution.

6. Answer: (b) Lowering Ea — Alternative path.

7. Answer: (c) More surface area — More exposed particles.

8. Answer: (b) Experiment — Rate law powers.

9. Answer: (c) Elementary step — Mechanism step.

10. Answer: (a) Independent of concentration — Rate = k.

11. Answer: (c) Linear — Integrated rate law.

12. Answer: (d) Slowest step — Controls overall rate.

13. Answer: (d) Increasing concentration — More collisions.

14. Answer: (d) Higher energy — More high-energy molecules.

15. Answer: (b) Chemically unchanged overall — Regenerated.

16. Answer: (a) k[A]^n — General rate law.

17. Answer: (c) Zero or fraction — Count of species.

18. Answer: (b) Zero, integer or fraction — Experimental.

19. Answer: (c) $AgNO_3 + NaCl$ — Ionic precipitation fast.

20. Answer: (d) Rusting — Oxidation slow.

21. Answer: (b) Concentration — More particles per volume.

22. Answer: (d) Effective collisions — Proper alignment needed.

23. Answer: (c) Constant — $t_{½} = 0.693/k$.

24. Answer: (b) mol dm^{-3} s^{-1} — Concentration/time.

25. Answer: (d) High-energy transition state — Peak of profile.

26. Answer: (d) Heterogeneous reactions — Contact area.

27. Answer: (d) Order — Dimensional analysis.

28. Answer: (c) Increases greatly — More effective collisions.

29. Answer: (b) Experimental rate law — Valid mechanism.

30. Answer: (d) Formed then consumed — Step species.

31. Answer: (b) Effective collisions — Energy and orientation.

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32. Answer: (b) E >= Ea and proper orientation — Successful collision.

33. Answer: (a) Minimum energy for reaction — Energy barrier.

34. Answer: (b) mol dm^{-3} s^{-1} — Concentration per time.

35. Answer: (b) Time interval — Δ concentration/Δ time.

36. Answer: (b) Tangent slope — At one time.

37. Answer: (a) Collision frequency — More particles per volume.

38. Answer: (d) Fraction above Ea — Boltzmann distribution.

39. Answer: (b) Surface area — More exposed particles.

40. Answer: (d) Alternative lower Ea path — Not consumed overall.

41. Answer: (d) Flattens and shifts right — More high-energy molecules.

42. Answer: (d) Experiment — Empirical orders.

43. Answer: (d) Temperature — k is T-dependent.

44. Answer: (b) Sum of powers — Rate law exponents.

45. Answer: (a) Independent of concentration — rate=k.

46. Answer: (d) Doubles rate — rate=k[A].

47. Answer: (b) Quadruples rate — rate=$k[A]^2$.

48. Answer: (c) Particles in elementary step — Whole number.

49. Answer: (d) Slowest step — Limits overall rate.

50. Answer: (b) Constant — $t_{½}$=0.693/k.

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